

L1: Algorithmic trading basics and financial markets

Course Code: COMP7415

Agenda

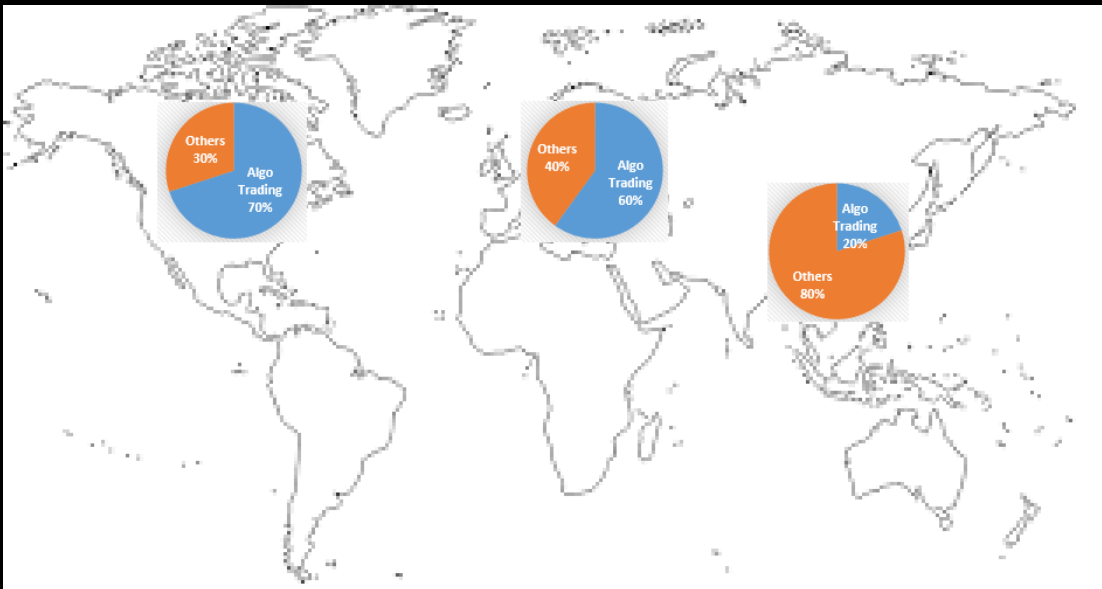
- The basic concept of algo-trading
- The market trend of algo-trading
- Comparison of different trading approaches
- Infrastructure needed to deploy algo-trading programs
- Trading mechanism and order matching
- Market jargon and terminology

What is algo-trading?

- Algorithmic trading, also called Algo Trading, Quant Trading, Robo-Trading, Program Trading
- Use of mathematical models and computer algorithms/programs to
 - generate trading signals (i.e. **Decision making**)
 - automate trading process (i.e. **Execution**)
- Objectives:
 - maximize profits
 - control execution costs
 - hedging and manage investment risks

Market Trend for Algo-Trading

- Expected to grow at CAGR 12.7% to US\$32b by 2028
- Largest market in North America
- Fastest growing in Asia Pacific



Reference:

<https://www.mordorintelligence.com/industry-reports/algorithmic-trading-market>

<https://www.prnewswire.com/in/news-releases/algorithmic-trading-market-to-reach-usd-31-494-million-by-2028-at-a-cagr-of-12-7-valuation-reports-897347529.html>

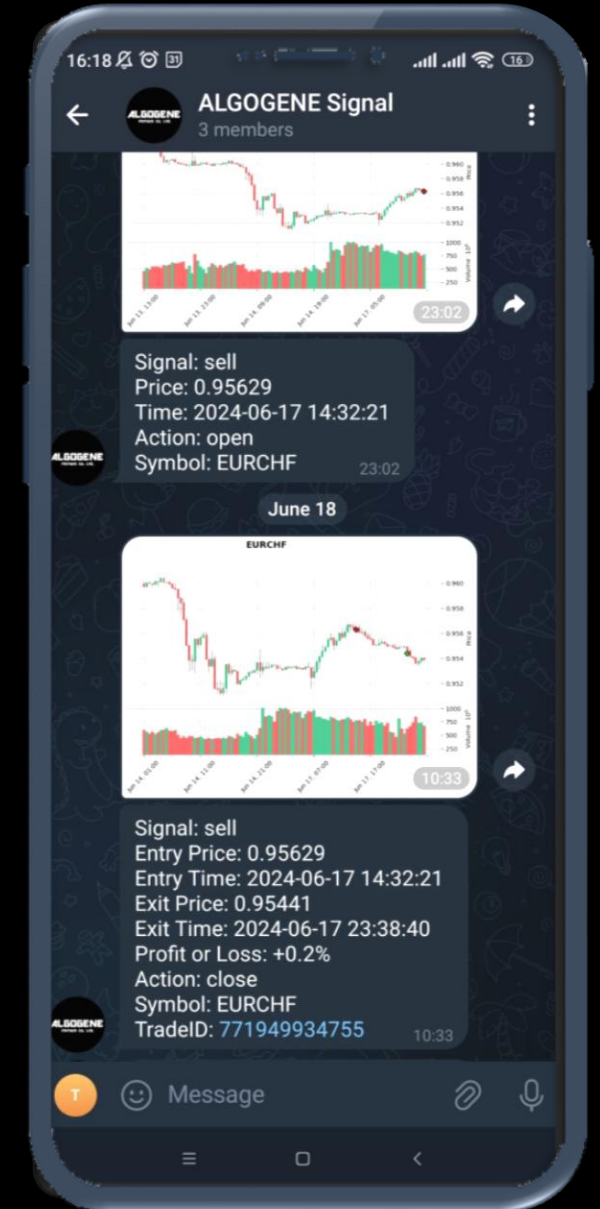
Market Trend for Algo-Trading

Algorithmic trading strategies account for approximately:

- 60% of all US equity volume
- 40% of all European equity volume
- 25% of all Forex transactions
- 20% of all US option trades

Robo-Trading vs Robo-Advisory

	Robo-Trading	Robo-Advisory
Decision making	✓	✓
Execution	✓	✗
Automation	Fully automated	Semi-automated
Outputs	Execution of trading signals	• Stock advices • Market summary • Risk alerts



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FinTech Co. Ltd.

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Warning: Beware of scam Telegram groups. Please note
that the only official group is @allogene_signal

Comparison of Trading Approaches

	Human Trading	Robo-Advisory	Robo-Trading
Working Hours	~9*5	~9*5	24*7
Execution Speed	Slow	Moderate	Immediately
Data Inputs	Limited	Almost unlimited	Almost unlimited
Trading Frequency	Low	Low - mid	Low – High
Scalability	Limited investment size due to stress	Moderate	High
Accuracy/ Discipline	Variable especially when you lose	Consistent	High
Customization	High	Medium	Low
User Control	Full	Partial	Minimal
Risk Management	Subjective	Algorithmic	Algorithmic

What should you consider to
implement a trading algo?

Algo is the vehicle. Data is the Fuel!

- **Data collection** – time and effort demanding in data sourcing and data cleaning
- **Data storage** – high volume of unstructured data (Numeric, Text, Image, Audio, Video)
- **Data management** – difficult to protect and manage through traditional databases due to inconsistent file and data formats
- **Data processing** – feasibility and efficiency

Other factors for infrastructure setup

- Real-time datafeed
- Historical data
- Computing hardware
- Exchange/broker connectivity
- Network and latencies
- Order routing system
- Order management system
- Algo maintenance
- Cybersecurity
- ...

General Trading Overview

Types of Trading

Fundamental:

- Stock Selection
- Ratio Analysis
- Sector Analysis
- Corporate and Executive Management

Technical:

- Chart pattern
- Trend Analysis

Quantitative:

- Rule based
- Econometric Forecasting
- Statistical Arbitrage
- NLP, sentiment analysis
- Machine learning

Types of Trading

Time Frames:

- Long Term: Months to Years
- Short Term: Days, Weeks, Months
- Intraday: Seconds to Hours
- High frequency: Fractions of Seconds

How trading works?

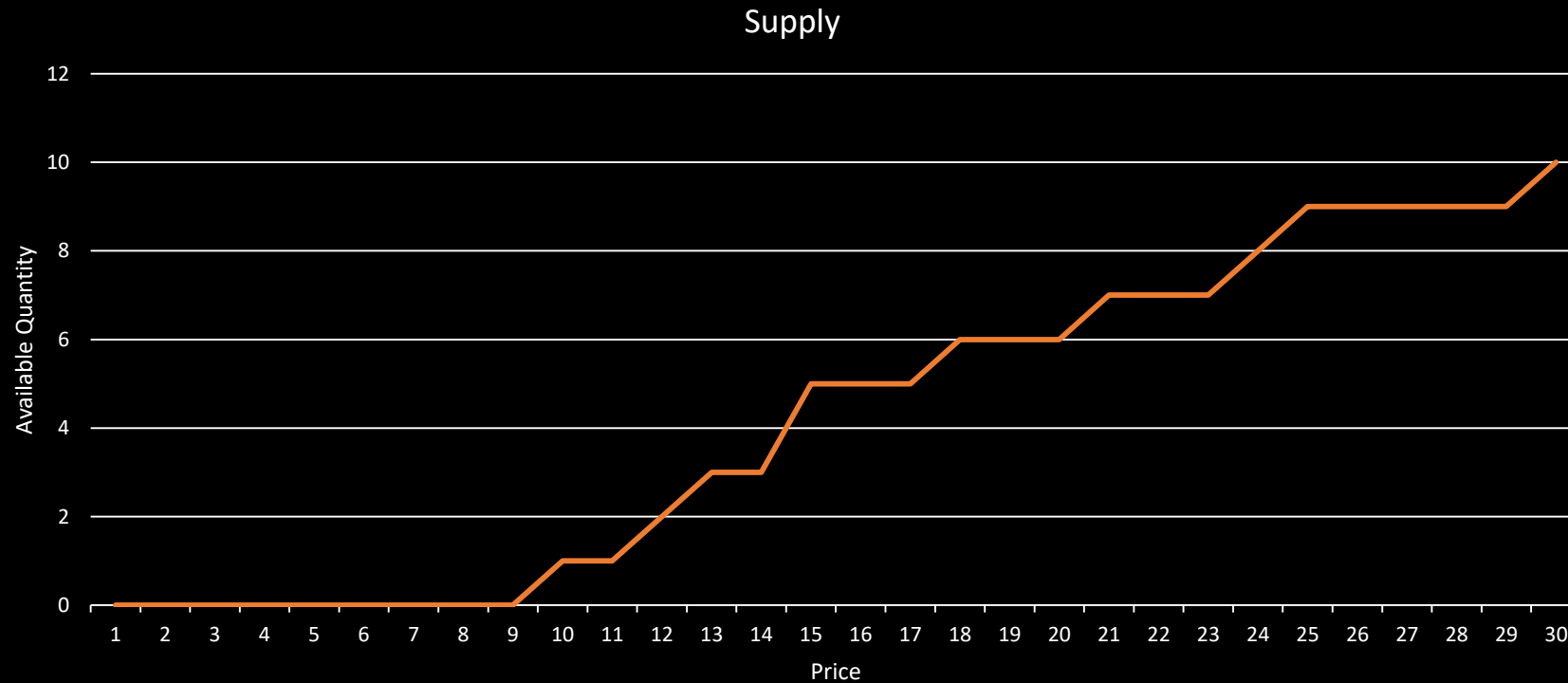
Supply

Current Stock holders
(each person holds 1 share)



Price they would be willing to take for their shares

Current Stock holders (each person holds 1 share)



Price	Supply
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	1
11	1
12	2
13	3
14	3
15	5
16	5
17	5
18	6
19	6
20	6
21	7
22	7
23	7
24	8
25	9
26	9
27	9
28	9
29	9
30	10

Demand

Potential Buyers
(each person wants to buy 1 share)



Price they would be willing to pay for a share

Potential Buyers

(each person wants to buy 1 share)



\$5



\$6



\$8



\$10



\$11



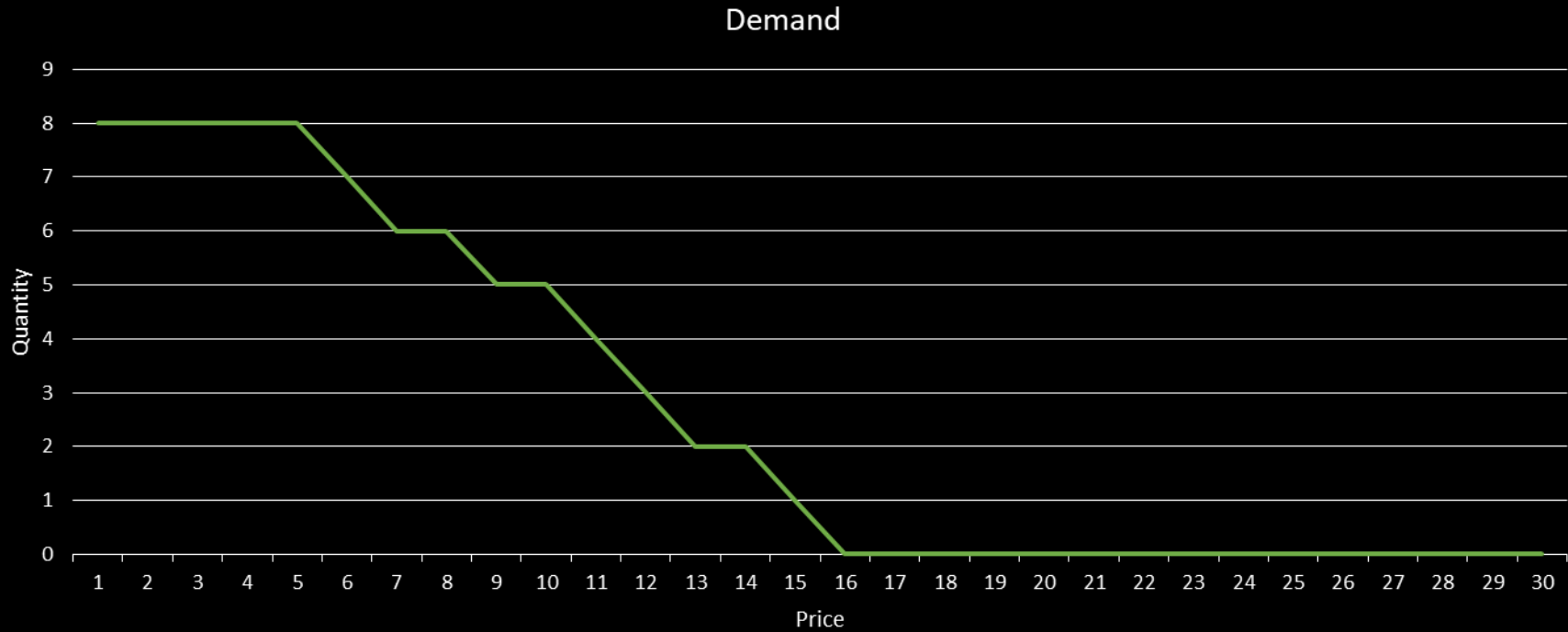
\$12



\$14



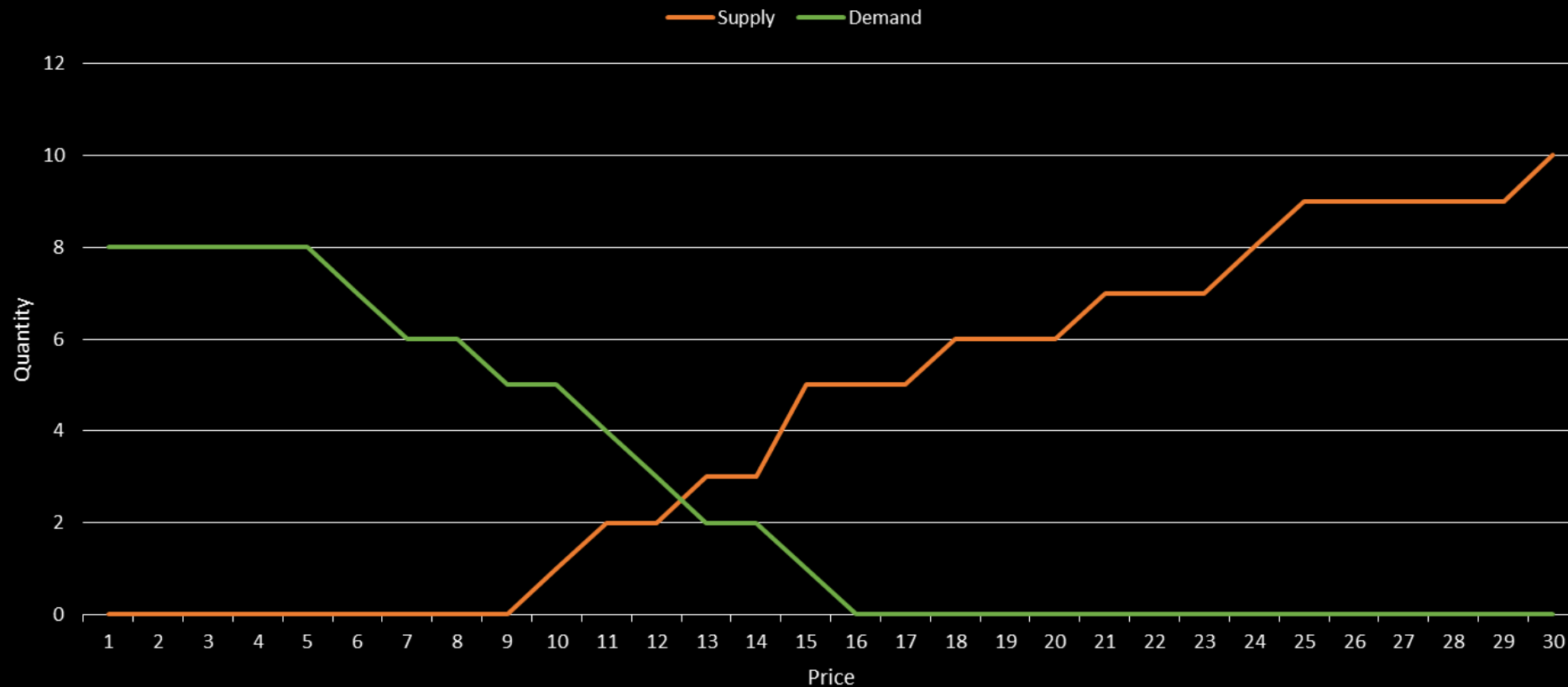
\$15



Price	Demand
1	8
2	8
3	8
4	8
5	8
6	7
7	6
8	6
9	5
10	5
11	4
12	3
13	2
14	2
15	1
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	0
26	0
27	0
28	0
29	0
30	0

Market Equilibrium

- The market is balanced at \$12 and quantity 2



Price	Supply	Demand
1	0	8
2	0	8
3	0	8
4	0	8
5	0	8
6	0	7
7	0	6
8	0	6
9	0	5
10	1	5
11	1	4
12	2	3
13	3	2
14	3	2
15	5	1
16	5	0
17	5	0
18	6	0
19	6	0
20	6	0
21	7	0
22	7	0
23	7	0
24	8	0
25	9	0
26	9	0
27	9	0
28	9	0
29	9	0
30	10	0

Order Matching

Current Stock holders
(each person holds 1 share)



Potential Buyers

(each person wants to buy 1 share)

After Order Matching



Market Terminology

Common market terminology

- Long and short position
- Short-selling
- Order book
- Bid price, ask price
- Market spread
- Order types
- Slippage
- Mark-to-market
- Trading system

Understanding Long and Short Positions

- Long Position
 - **Definition:** Buying an asset with the expectation that its price will rise.
 - **Goal:** Sell at a higher price to realize a profit.
 - **Example:** Buying 100 shares of a stock at \$50, selling when it reaches \$70.



Understanding Long and Short Positions

- Short Position

- **Definition:** Selling an asset that you do not own, with the intention of buying it back later at a lower price.
- **Goal:** Profit from a decline in the asset's price.
- **Example:** Selling 100 shares of a stock at \$70, buying back at \$50.



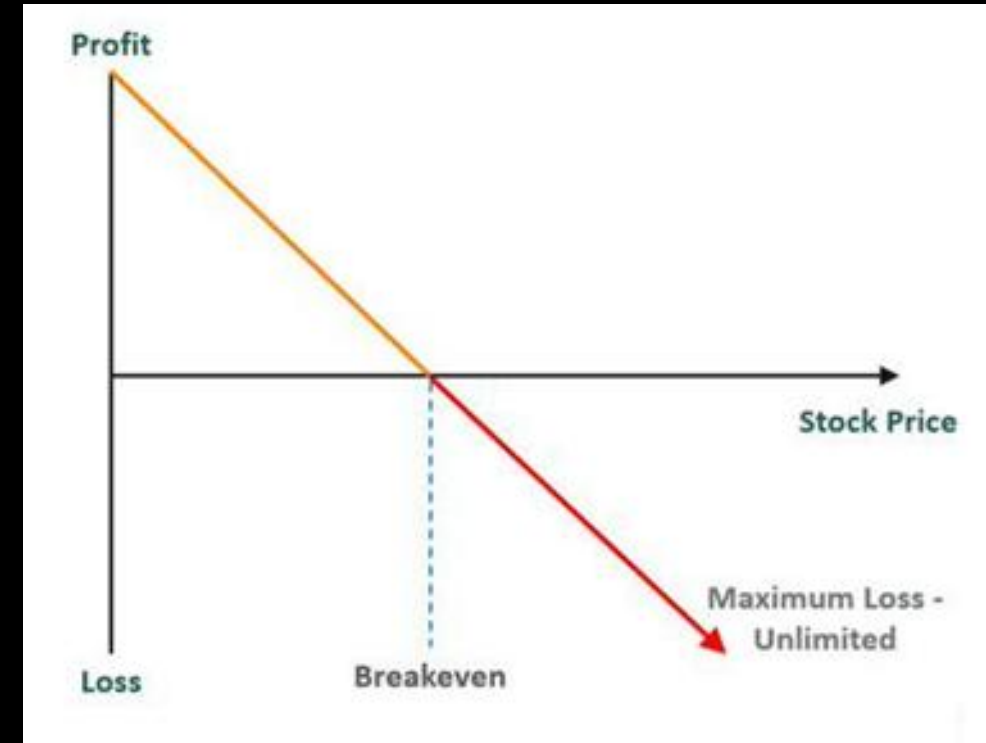
How Short-Selling Works

1. Borrowing Shares: Obtain shares from a broker to sell.
2. Selling the Borrowed Shares: Sell the shares at the current market price.
3. Buying Back Shares: Later, buy the same number of shares at a lower price.
4. Returning Shares: Return the borrowed shares to the broker.



Risks of Short-Selling

- **Unlimited Loss Potential:** If the asset price rises significantly.
- **Margin Calls:** Brokers may require additional funds if the trade goes against you.
- **Regulatory Risks:** Short-selling can be subject to restrictions.



Order Book

- The Order Book is a real-time, continually updating list of buy and sell orders for a specific financial instrument, sorted by price level.
- Key components
 - Ask Book: the list/queue for sellers
 - Bid Book: the list/queue for buyers
 - Price: The price points at which investors are willing to trade
 - Size: The number of shares, contracts, or lots that traders want to buy or sell

How does Order Book present?

1. Order Ledger
2. Top of the Book
3. Cumulative Book

Order Ledger

- The order book is listed in a sorted price ledger, which can be ascending or descending.
- The longer the price ledger is, the more liquidity an instrument provides. It is also called "market depth".



Top of the Book

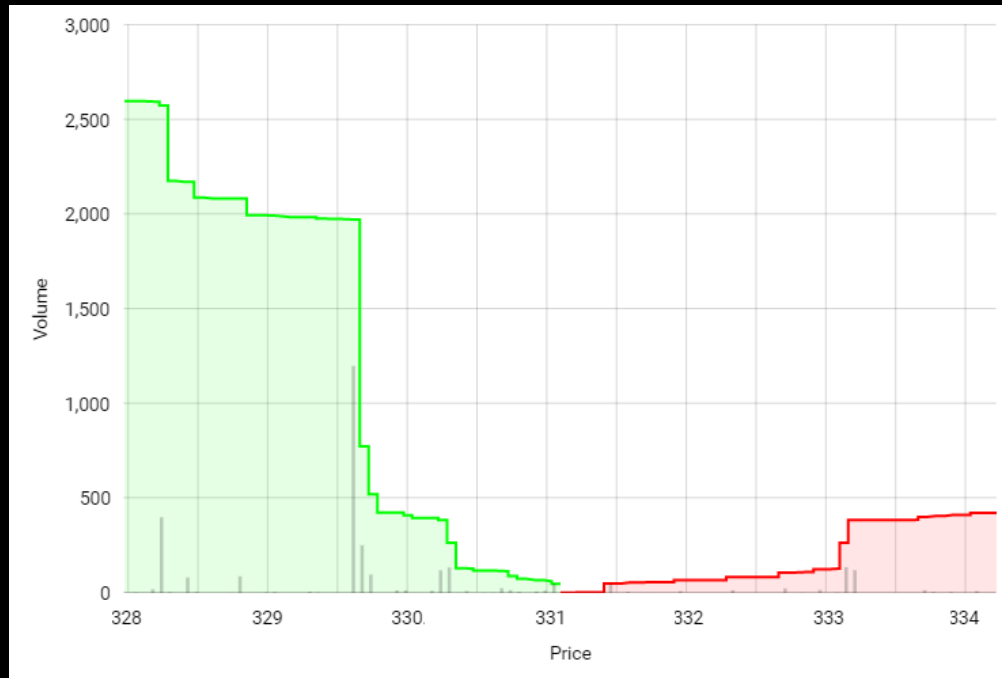
- The order book is separated into bid-book and ask-book
- Bid orders are sorted in descending price
- Ask orders are sorted in ascending price
- The highest bid and the lowest ask are referred to "the top of the book"

Order Book

	Bid	Ask	
3,151	9139.60	9152.80	167
1,574	9138.40	9153.21	1,600
338	9138.20	9153.60	736
789	9138.10	9153.71	1,602
800	9138.00	9154.10	1,649
309	9137.79	9154.21	1,965
2,140	9137.60	9154.50	3,984
48	9137.40	9154.60	763
144	9137.29	9154.71	157
136	9137.20	9154.80	56
528	9137.00	9155.50	320
1,654	9136.90	9155.71	200
40	9136.79	9156.10	16
46	9136.70	9156.21	219
608	9136.60	9156.30	40
48	9136.50	9156.40	80

Cumulative Order Book

- For each of the bid and ask book, order size accumulates from the top of the book.



Order Book		Market Trades	
		0.1 ▼	
Price(USDT)	Qty(BTC)	Total(BTC)	
61,284.8	11.972	54.521	
61,284.7	7.282	42.549	
61,284.0	2.529	35.267	
61,283.9	1.305	32.738	
61,283.8	1.795	31.433	
61,283.7	4.569	29.638	
61,283.5	5.773	25.069	
61,283.4	1.795	19.296	
61,283.3	3.590	17.501	
61,283.2	13.911	13.911	
61,284.4 ↑ 61,291.9			
61,282.3	12.075	12.075	
61,282.2	0.816	12.891	
61,282.1	2.693	15.584	
61,282.0	4.712	20.296	
61,281.9	9.444	29.740	
61,281.8	1.795	31.535	
61,281.7	3.672	35.207	
61,281.6	5.181	40.388	
61,278.4	15.301	55.689	
61,267.2	15.306	70.995	



Summary

News

Chart

Conversations

Statistics

Historical Data

Profile

Financials

Analysis

Options

Holders

Sustainability

NasdaqGS - Nasdaq Real Time Price • USD

Apple Inc. (AAPL)

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207.49 -2.19 (-1.04%) 207.86 +0.37 (+0.18%)

At close: June 21 at 4:00 PM EDT

Pre-Market: 7:41 AM EDT

Start Trading >>

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1D 5D 3M 6M YTD 1Y 5Y All

Key Events

Mountain ▾

Advanced Chart



Volume Not Available

209.68

208.00

207.84

207.50

207.00

Previous Close	209.68	Day's Range	207.11 - 211.89	Market Cap (intraday)	3.182T	Earnings Date	Aug 1, 2024 - Aug 5, 2024
Open	210.39	52 Week Range	164.08 - 220.20	Beta (5Y Monthly)	1.25	Forward Dividend & Yield	1.00 (0.48%)
Bid	207.01 x 3000	Volume	246,421,353	PE Ratio (TTM)	32.22	Ex-Dividend Date	May 10, 2024
Ask	209.94 x 200	Avg. Volume	67,794,775	EPS (TTM)	6.44	1y Target Est	207.58

Bid Price & Bid Size

- Bid price is the highest price that a buyer (bidder) is willing to pay for a particular security.
- Bid size represents the quantity a buyer is willing to purchase at the bid price.
- The top price and size in a bid order book.
- For example, if a buyer bids \$207.01 x 3000 for a stock, it means they are willing to buy 3000 shares for not more than \$207.01

Bid

207.01 x 3000

Ask Price & Ask Size

- Ask price is the lowest price a seller is willing to accept for a security.
- Ask size represents the quantity a seller is willing to sell at the ask price.
- The top price and size in an ask order book.
- For example, if a seller asks \$209.94 x 200 for a stock, it implies they are willing to sell 200 shares for not less than \$209.94

Ask

209.94 x 200

HKEX Market Data Pricing

- Level 1 data:
 - bid/ask price: HK\$500 /month
- Level 2 data:
 - full order book: HK\$5,000 /month

Reference: <https://www.hkex.com.hk/eng/ods/historicalData.aspx>

HKEX 香港交易所				
Historical Full Book - Securities Market (Binary Format)	Daily		\$5,000/ Month	
Provides information on every order and trade of Main Board and GEM stocks in binary format. File is available on a daily basis, normally at around 8:30pm.				
Historical Full Book - Securities Market (CSV Format)	Daily		\$5,000/ Month	
Provides information on every order and trade of Main Board and GEM stocks in CSV format. File is available on a daily basis, normally at around 10:00pm.				
Historical Order Book and Statistics Update - Securities Market	Daily		\$2,500/ Month	
Provides intra-day order book and statistics information recorded on the HKEX's Main Board and GEM stocks in CSV format. File is available on a daily basis, normally at around 10:00pm.				
Bid and Ask Record - Main Board & GEM (Daily File)	Daily		\$500/ Month	
Provides intra-day bid and ask information on Main Board and GEM stocks				
Bid and Ask Record - Main Board & GEM (Monthly File)	Monthly		\$500/ Month	
Provides intra-day bid and ask information on Main Board and GEM stocks				
Trade File - Securities Market (Binary Format)	Daily		\$2,500/ Month	
Provides trade details, including trade time, price and quantity, of each transaction recorded on the Main Board and GEM in binary format. File is available on a daily basis, normally at around 8:30pm.				
Trade File - Securities Market (CSV)	Daily		\$2,500/ Month	
Provides trade details, including trade time, price and quantity, of each transaction recorded on the Main Board and GEM in CSV format. File is available on a daily basis, normally at around 10:00pm.				

Market Spread

- The difference between the bid price and the ask price is known as the bid-ask spread.
- The bid-ask spread is a key measure of the liquidity of an asset or security.
- A smaller spread indicates a more liquid market, while a larger spread indicates a less liquid market.
- For example, if the bid price for a stock is \$24 and the ask price is \$25, the bid-ask spread will be \$1.

Percentage Spread

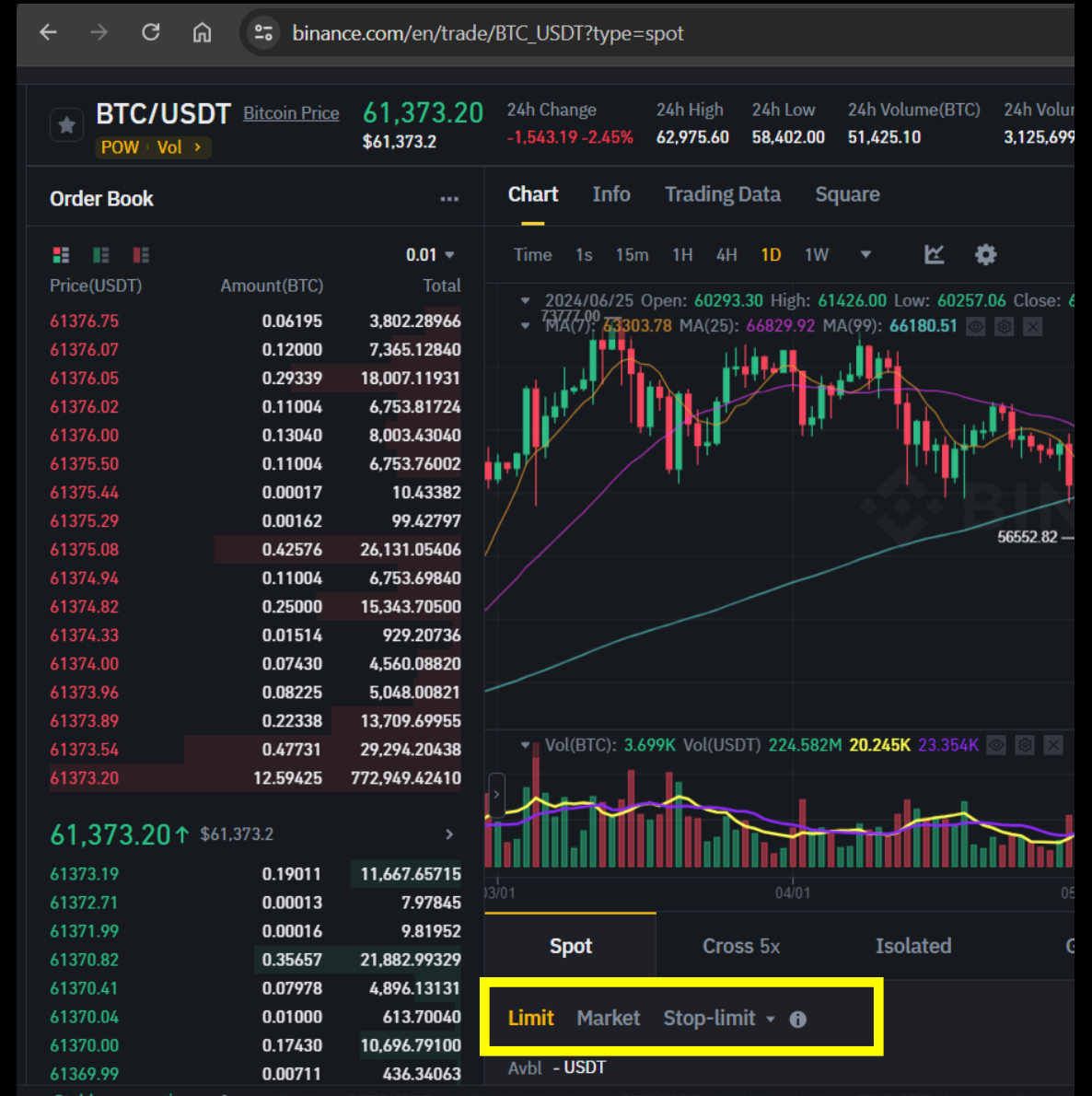
- Percentage spread is often used for comparing the spread between different instruments

$$\text{Percentage Spread} = \frac{\text{AskPrice} - \text{BidPrice}}{\text{MidPrice}} \times 100\% = \frac{2 \times (\text{AskPrice} - \text{BidPrice})}{\text{AskPrice} + \text{BidPrice}} \times 100\%$$

- For example, if the bid price for a stock is \$24 and the ask price is \$25,
 - the bid-ask spread will be $(25-24) = \$1$
 - the percentage spread will be $2 \times (25-20)/(25+20) = 4.08\%$

Order Types

1. Market Order
2. Limit Order
3. Stop Order



Market Order

- A market order is to buy or sell at the best available price in the current market.
- That is, when opening a buy (sell) position using market order, it will try to execute at the best ask (bid) price.
- A market order typically ensures an execution, but it does not guarantee a specified price. It is appropriate to use market orders when you want an immediate execution.

The screenshot displays a trading interface with two side-by-side order forms. The top navigation bar includes 'Limit', 'Market' (highlighted in orange), and 'Stop-limit' with a dropdown arrow and an information icon. On the right, there are links for 'Transfer', 'Auto-Invest', and 'Buy with EUR'.

Left Form (Buy BTC):

- Avbl: 0.00359157 USDT
- Price input field with 'Market' and 'USDT' dropdowns.
- Total input field with 'USDT' dropdown.
- Max Buy: 0.00000 BTC
- Est. Fee
- Buy BTC button (green)

Right Form (Sell BTC):

- Avbl: 0.00000674 BTC
- Price input field with 'Market' and 'USDT' dropdowns.
- Amount input field with 'BTC' dropdown.
- Max Sell: 0.4116156 USDT
- Est. Fee
- Sell BTC button (red)

Limit Order

- A limit order is to buy or sell with a condition on the maximum price to pay or the minimum price to receive (the "limit price").
- If the order is filled, it will only be at the specified limit price or better. However, there is no guarantee of execution.
- A limit order may be appropriate when you think you can buy at a price lower than (or sell at a price higher than) the current market quote.

Limit Order Example



Limit Order Example

- The last trade price is roughly \$138
 - investor who wants to buy (or sell) immediately would place a market order, which would be executed at or near the current price of \$138 (white line) provided that the market was open.
 - investor who wants to buy the stock when it dropped to \$131.78 would place a buy limit order with a limit price of \$131.78 (green line). If the price falls to \$131.78 or lower, the limit order would be triggered and executed at \$131.78 or below. If the stock doesn't drop to \$131.78 or below, no execution would occur.
 - investor who wants to sell the stock when it reached \$143.82 would place a sell limit order with a limit price of \$143.82 (red line). If the price rises to \$143.82 or higher, the limit order would be triggered and executed at \$143.82 or above. If the stock doesn't rise to \$143.82 or above, no execution would occur.

Stop Order

- A stop order is to buy or sell at the market price when it has traded at or through a specified price (the "stop price").
- If the stock reaches the stop price, the order becomes a market order and is filled at the next available market price. If it doesn't reach the stop price, the order will not be executed.
- A stop order may be appropriate in these scenarios:
 - you want to buy when a stock breaks out above a certain level, and believe that it will continue the trend
 - your holding stock has risen a lot and you want to protect the gain when it begins to fall

Stop Order Example



Stop Order Example

- The stop buy order will trigger when the price reaches 143.82 or above, and will execute as a market order at that current price.
- Thus, if the price rise further after hitting the stop price, it is possible that the order could be executed at a price higher than the stop price.
- Similarly for the stop sell order, once the stop price of \$131.78 is reached, the order could be executed at a lower price

Slippage

- Slippage refers to the difference between the expected execution price and the price it actually traded
- It often occurs
 - during periods of higher volatility when market orders are used
 - when large orders are executed when market depth is not sufficient to maintain the expected price of trade

Slippage Example

- Let's assume you have an algorithm set to buy a stock when the price drops to \$50. The algorithm detects the price drop and places a buy order.
- However, due to high demand or low liquidity, the order gets executed at \$51. This \$1 difference is the slippage.
- This means that even if you were expecting to spend \$5000 on 100 shares, you would end up spending \$5100. The \$100 is the cost of slippage

Profit/Loss Calculation

- **Entry:** Based on a trade signal, generate order to go short or long a certain financial instrument in a certain quantity. Trade results in a certain position in this security
- **Mark-to-Market:** As the price of the security changes, so does your **unrealized PnL**

*For long order, $PnL = Quantity * (P_{bid} - P_{entry})$*

*For short order, $PnL = -1 * Quantity * (P_{ask} - P_{entry})$*

- **Exit:** Generate order to exit the position and create a **realized PnL**

$PnL = Side * Quantity * (P_{exit} - P_{entry})$

Mark-to-Market

- It is an accounting method to calculate the asset or portfolio value according to its current market value, rather than its book value
- It involves determining the price at which you could immediately sell an asset or close a position
- Example:
 - Suppose you purchase 100 shares of ABC stock
 - The current bid and ask price of ABC are \$9.5 and \$10.5
 - Mark-to-market value of your stock will be $100 * \$9.5 = \9500

Order Management System (OMS)

1. Order based system

- Transactions are managed in a round order manner
- Partial close is not supported
- Trading platforms
 - MetaTrader, TradingView, MetaStock, etc

2. Position based system

- Transactions are independent and no linkage with each other
- Partial close is allowed
- Trading Platforms
 - Most of the banks (eg. HSBC, SBC, BOC, etc)
 - Interactive Brokers, Binance, etc

Position Based vs Order Based System

Suppose you open 3 offsetting trades:

#1: buy 2 shares of ABC at \$123

#2: sell 1 share of ABC at \$124

#3: sell 1 share of ABC at \$125

For order based system, the **unrealized** PnL will be $2 * (\text{bid} - 123) + 1 * (124 - \text{ask}) + 1 * (125 - \text{ask}) = \$3 + 2 * (\text{bid} - \text{ask})$. You need to close all the 3 trades to realize the PnL.

For position based system, the orders will be offsetted based on FIFO, and PL will become realized so long as the position is net to zero. Thus, your PnL will be **realized** at \$3.

Key Takeaways

- Learn the basic concept of algo-trading and its market trend
- Understand the pros and cons of different trading approaches
- Know about the infrastructure/resources needed to develop and deploy algo-trading programs in real market
- Understand the trading mechanism and order matching in financial market
- Equip the knowledge about different market terminology (eg. Order book, order types, market spread, slippage, order/position based trading system, etc)